

Lab 5 Reflection

In this lab, I learned about several important steps that are used to prepare data before it can be used for machine learning. One of the main things I learned was how to handle missing values. The Titanic dataset had missing values in the Age column, so we used the median to replace them. I learned that the median can be better than the mean when the data has unusually high or low values because the median is not affected as much by extreme values. This gives us a reasonable value to use without removing entire rows of data.

I also learned about One-Hot Encoding and why it is useful. Some of the Titanic data was stored as text, such as whether a passenger was male or female and where they boarded the ship. Machine learning models generally work with numbers, so One-Hot Encoding changes these categories into numerical columns containing 0s and 1s. I thought this was interesting because it allows the model to work with categories without treating them as if they have a numerical order. Using `drop_first=True` also removes one unnecessary category because the model can determine it from the other columns.

The last major thing I learned was Feature Scaling. The Age and Fare columns had very different ranges, so we used `StandardScaler` to put them on a more similar scale. This helps prevent differences in the size of the numbers from causing problems for models that are affected by feature scale. I also learned that scaling is not always necessary. For example, Decision Trees does not usually need feature scaling because they make decisions based on splits in the data rather than the overall size of the values.

REFLECTION JOURNAL – Module 05: Data Preparation

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My Learning Experience

This lab helped me understand that data preparation is an important part of the machine learning process. Working with the Titanic dataset showed me that raw data can contain missing values, categorical information, and numerical variables with different scales. Before a model can use the data effectively, these issues need to be addressed.

Missing Values and Imputation

I learned how **missing-value imputation** can be used instead of deleting rows and losing useful information. I also understood why the median was used instead of the mean.

One-Hot Encoding

I learned that machine learning models generally need numerical data rather than text categories.

The Titanic dataset contained categorical columns such as Sex and Embarked. Using

`pd.get_dummies()`, I converted these categories into numerical columns containing 0 and 1.

I also learned that `drop_first=True` can be used to remove one of the encoded categories and avoid unnecessary duplication. This showed me that encoding is not simply changing text into numbers; it prepares categorical information in a format that can be used by a machine learning model.

Feature Scaling

I also learned that scaling should be applied to the appropriate numerical features rather than automatically applying it to every column. The encoded categorical variables should not be treated the same way as continuous numerical features.

Challenges and Problem-Solving

I encountered a few challenges during the lab. Initially, I received a **SyntaxError** because some of the Python statements were combined incorrectly. I corrected the code by separating the statements and running the cell again. Later, I received a `NameError` stating that `pd` was not defined. I learned that the setup and import cells must be executed so that Python recognizes the libraries and variables being used. After running the setup cell again, I was able to continue.

I also received a pandas warning about using `inplace=True`. Although it did not stop the program from running, it helped me understand that warnings can provide useful information about future changes in programming libraries. These challenges taught me to read error messages carefully and solve problems step by step rather than becoming discouraged.

Understanding the Complete Workflow

The biggest lesson from this lab was seeing how the different data-preparation steps connect. First, I loaded and examined the dataset. Next, I handled missing values through imputation. I then converted categorical variables into numerical features using One-Hot Encoding. Finally, I applied Feature Scaling to the selected numerical variables.

I now understand that data preparation is more than simply cleaning a dataset. Each step prepares the data for the next stage of the machine learning workflow. Overall, this lab gave me a better understanding of why preprocessing techniques are needed and how they work together to make data more suitable for machine learning.

Reflection

In this "Data Preparation" experiment, I learned more about an important part of the machine learning workflow: data preprocessing. Before this lab, I knew that data needed to be cleaned before using it to train a model, but I did not fully understand why each step was important. After working with the Titanic dataset, I better understood how missing values, categorical data, and different feature scales can affect machine learning results.

First, I learned how to handle missing values. For the "Age" feature, I used the median to fill in the missing values. I learned that the median can be a better choice when the data has outliers or is not evenly distributed. The mean can be affected by very high or very low values, while the median is less affected by them. This helped me understand that there is not always one method that works for every dataset. We need to look at the data before deciding how to handle missing values.

Second, I learned why categorical data needs to be changed into numerical data. One Hot Encoding changed categories into columns with 0s and 1s so that the machine learning model could use them. I also learned about using `drop_first=True`. This removes one of the extra categories and can help avoid multicollinearity. Before this lab, I knew One Hot Encoding was used to change text into numbers, but I did not understand clearly why dropping one category could be useful.

Another important thing I learned was that different machine learning models may need different types of preprocessing. For example, models such as Logistic Regression, SVM, and KNN can be affected by the scale of the features. Decision Trees do not need feature scaling

because they make decisions using thresholds. This showed me that data preprocessing is not just a fixed list of steps that should always be applied in the same way.

Overall, this lab helped me understand that data preparation is more than just cleaning a dataset. Each preprocessing choice should be based on the type of data and the model being used. I also realized that small decisions made before training a model can affect the final results. This lab gave me a better understanding of how the different steps in a machine learning workflow connect with each other.